



CIRCUIT MASTER

An Educational Electronics Game

GAME INSTRUCTIONS

2 Players

Ages 14+

30-45 min

QUICK TIPS

- 1 No wiring needed — rail connections are automatic on both sides of the board.
- 2 Place Resistors/Capacitors directly to build series or parallel circuits.
- 3 Match the Objective Card value to score points on your turn.
- 4 Higher-tier objectives (more components) are worth more points.

OBJECTIVE

Build circuits on the shared breadboard to match Objective Cards and score the most points.

COMPONENTS

- 1 × A4 Game Board
- 40 × 100 Ω Resistor cards
- 20 × 200 Ω Resistor cards
- 20 × 300 Ω Resistor cards
- 40 × 10 μ F Capacitor cards
- 20 × 20 μ F Capacitor cards
- 20 × 30 μ F Capacitor cards
- 32 × Objective cards

SETUP

- 1 Place the board between both players.
- 2 Shuffle Component cards; deal each player 15 cards as their starting hand.
- 3 Shuffle Objective cards. Each player draws 3 Objective cards face-up.
- 4 Decide who goes first — Player 1 begins.

HOW TO PLAY

YOUR TURN

On your turn, perform up to 3 Actions:

PLACE

Play a card from your hand and place the component on a valid hole on the board.

DRAW

Take 1 card from the shared Component deck.

COMPLETE

Finalize your circuit: the board is checked against your Objective Cards and you score any that match.

OBJECTIVE TIERS

SETUP — 2 components

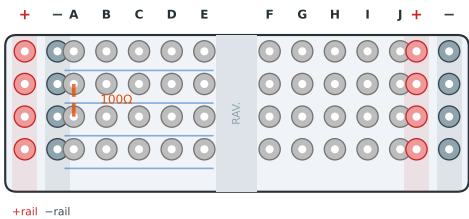
Simple series or parallel objectives. Rail connections are automatic — no wiring needed, just place Resistors/Capacitors directly to build a circuit.

BUILD — 3 components, EXPERT — 4 components

Place more Resistors and Capacitors to hit higher-tier Objective Card values for more points. All tiers are available from the start.

THE BREADBOARD

- Holes A-E in the same row are internally connected (left half).
- Holes F-J in the same row are internally connected (right half).
- The Ravine divides the board into two independent halves.
- Each half has its own + rail and – rail — you can work on either side.
- A valid circuit must connect a + rail to a – rail through your components.



Series

RESISTANCE

$$R = R_1 + R_2$$

CAPACITANCE

$$1/C = 1/C_1 + 1/C_2$$

Parallel

RESISTANCE

$$R = R_1 \times R_2 / (R_1 + R_2)$$

CAPACITANCE

$$C = C_1 + C_2$$

► Resistors in parallel → lower resistance.

► Capacitors in series → lower capacitance.

EXAMPLES

100 Ω + 200 Ω in series → 300 Ω

100 Ω || 100 Ω in parallel → 50 Ω

10 μF + 10 μF in series → 5 μF

10 μF || 20 μF in parallel → 30 μF

SCORING

- At the end of each of your turns, check your Objective Cards.
- If your components on the board match an Objective target → score its points.
- Discard the completed Objective and draw a new one from the deck.
- Only your own components count toward your objectives.

OBJECTIVE TIERS

SETUP

30-40 pts

2 components

BUILD

40-45 pts

3 components

EXPERT

65 pts

4 components

WINNING

After 6 turns each, the player with the most total points wins!
Tie-break: the player who completed more Objective Cards wins.